AMS 315: Computing Assignment 1 - Part A

Kyle Chan

Introduction: The goal of Part A is to clean and prepare raw data for analysis, a common first task for a statistician. I was given two data sets for Part A, one with IDs and IVs, and one with IDs and DVs. I was tasked to check for accuracy of data, which includes sorting the two files by ID, merging them together, reporting how much data is missing, and imputing the missing values so the full sample can be analyzed.

Methods: I used R to perform my statistical analysis. Both files were sorted by subject ID, and merged with the merge() function. A full outer join was used, so that a subject ID appearing in only one file would still be kept. No manual cut-and-paste was used. Missing values were counted with is.na(), and the missing-data pattern was examined with md.pattern() from the mice package.

Of the 569 subject IDs, 500 had an IV value and 520 had a DV value (12.1% and 8.6% missing, respectively); 454 had both, 566 had at least one, and 3 had neither. (Table 1, Appendix B). The 3 IDs with no data were removed, and the remaining missing values were imputed with MICE (norm.boot), which fills each gap using regression on bootstrap resamples, avoiding listwise deletion and mean/median imputation; with under 30% missing, the method choice should have little effect. A least-squares regression, $DV = \beta_0 + \beta_1 \cdot IV + \varepsilon$, was then fitted to the 566 completed records, with an ANOVA F-test and a 95% confidence interval for β_1.

Results: The fitted function is $\widehat{DV} = 30.62 + 4.87 \cdot IV$, with $R^2 = 0.717$. In the ANOVA (Table 2, Appendix B), $SS_{reg} = 13020.3$ (1 df) and $SS_{res} = 5139.4$ (564 df), giving $F(1, 564) = 1428.9$ and $p < 2.2 \times 10^{-16}$. The 95% confidence interval for the slope is $(4.617,\ 5.123)$; since it excludes zero, $H_0: \beta_1 = 0$ is rejected. (Figure 1, Appendix A) shows the scatterplot with the fitted line.

Conclusion: There is a strong, positive, statistically significant linear association between IV and DV. About 71.7% of the variation in DV is explained by the IV. Each one-unit increase in IV is associated with an estimated increase of about 4.87 units in DV. The narrow confidence interval shows the slope is estimated precisely. The estimated generating function is $\widehat{DV} = 30.62 + 4.87 \cdot IV$. The fraction of missing data was small (12.1% for IV, 8.6% for DV), well under the 30% threshold, so the imputation is unlikely to have changed these conclusions.

AMS 315: Computing Assignment 1 - Part B

Kyle Chan

Introduction: The objective of Part B is to recover the function that generated the dependent variable y from the independent variable x. The data file has 474 complete observations, each with a subject ID, an x value (range 1.008 to 2.000), and a y value (range 0.0157 to 0.0668). The analysis looks for a transformation that best fits the linear regression model, then checks the adequacy of the chosen model with an approximate lack-of-fit test.

Methods: R was used for all analysis. First, I fit a simple linear regression of y on x and looked at the scatterplot and the residual plot. Then, I tried different transformations: x, $\log x$, $1/x$, $\sqrt{x}$ and x^2 for the independent variable, and y, $\log y$, $1/y$, $\sqrt{y}$ and $1/\sqrt{y}$ for the dependent variable. I compared them using R^2 and the residual plots, and picked the model $1/y = \beta_0 + \beta_1 x + \varepsilon$. Since almost none of the x values repeated exactly, I grouped close x values into bins of width 0.05, which gave 19 bins with 17 to 45 points each. I replaced each x with the average x of its bin and then ran a lack-of-fit test using pureErrorAnova() from the alr3 package.

Results: The untransformed model had $R^2 = 0.5610$, and its residual plot (Figure 2) showed a funnel shape, meaning the variance was not constant. Of the transformations compared (Table 3, Appendix B), $1/y$ worked best, raising R^2 to 0.6602; x^2 with $1/y$ was slightly higher (0.6668), but I kept the simpler model since the gain was very small. After the transformation, the residual plot (Figure 3) showed a random band with roughly constant spread. The fitted model is $\widehat{1/y} = -1.4826 + 25.1132x$. In the ANOVA (Table 4, Appendix B), $SS_{reg} = 23918$ (1 df) and $SS_{res} = 12311$ (472 df), giving $F(1, 472) = 916.98$ and $p < 2.2 \times 10^{-16}$. The 95% CI for the slope is $(23.48,\ 26.74)$, which excludes zero, so I rejected $H_0: \beta_1 = 0$. The lack-of-fit test showed no significant lack of fit, $F(17, 455) = 1.34$, $p = 0.164$ (Table 5, Appendix B).

Conclusion**:** There is a strong, statistically significant relationship between x and y. After the reciprocal transformation, x explains about 66% of the variation in $1/y$, compared with 56% for the raw data. Solving for y, the estimated generating function is $\hat{y} = \frac{1}{-1.4826+25.1132x}$

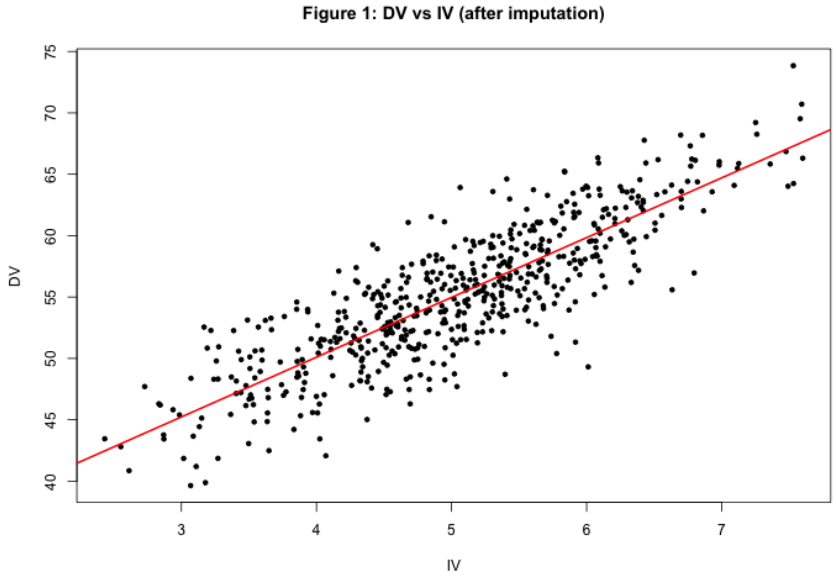
and the lack-of-fit test shows that a straight line is adequate for the transformed data. The intercept was not significant ($p = 0.247$), and its confidence interval $(-3.99,\ 1.03)$ contains zero. This suggests the true function may be closer to $y = \frac{1}{25x} = \frac{0.04}{x}$. The untransformed data also showed no significant lack of fit ($p = 0.548$), so the case for the transformation rests mainly on the higher R^2 and the improved residual plot, not on the lack-of-fit test.

AMS 315: Computing Assignment 1 - Appendix

Kyle Chan

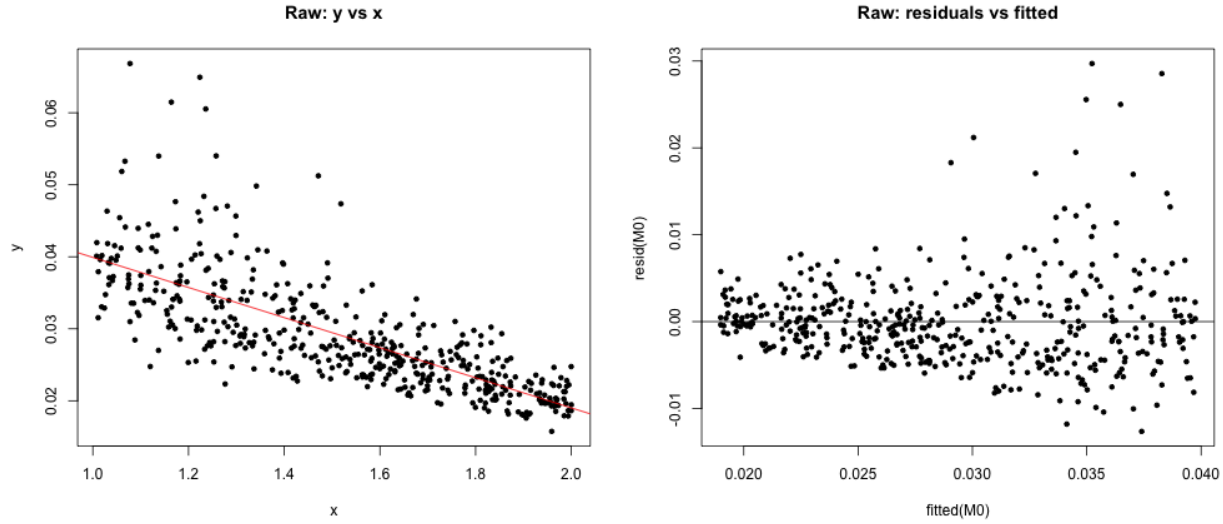
Appendix A: Figures

Figure 1. Scatterplot of DV vs IV with fitted regression line (Part A, after imputation)



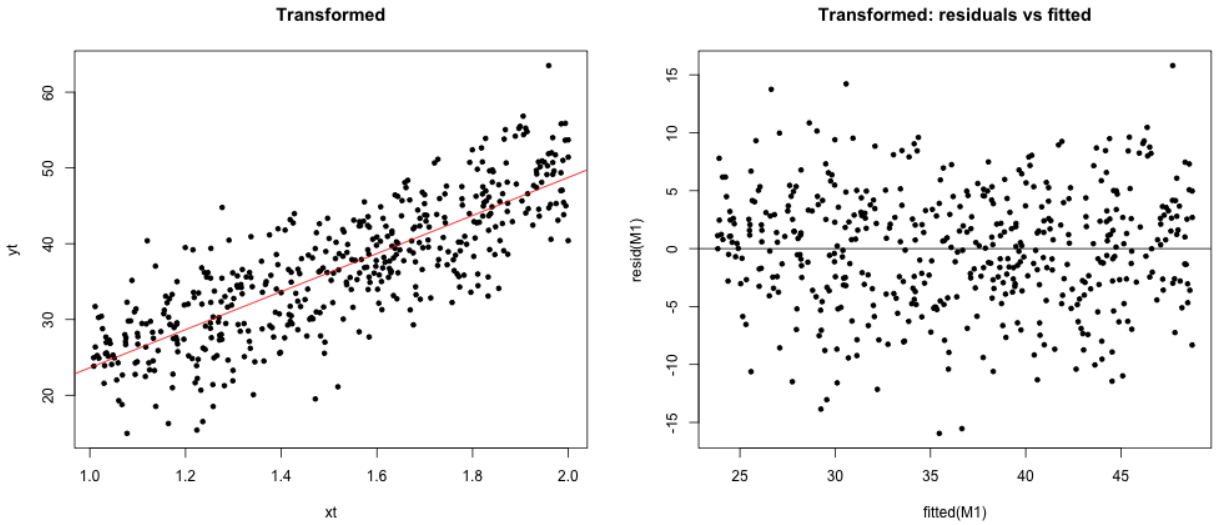
DV = 30.62 + 4.87·IV

Figure 2. Raw Data (Part B), scatterplot of y vs x and residuals vs fitted values



The scatterplot curves downward, and the residuals fan out as fitted values increase.

Figure 3. Transformed Data (Part B): Scatterplot of 1/y vs x and residuals vs fitted values



After the 1/y transformation, the relationship is closer to linear, and the residuals form a random band with constant spread.

Appendix B: Tables

Table: Table 1. Subject ID counts

Quantity	Value
Total subject IDs	569.0000
IDs with an IV value	500.0000
IDs with a DV value	520.0000
IDs with both IV and DV	454.0000
IDs with at least one of IV or DV	566.0000
IDs missing both	3.0000
Fraction of IV missing	0.1213
Fraction of DV missing	0.0861

Table: Table 2. ANOVA, Part A

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
IV	1	13020.35	13020.35	1428.86	0
Residuals	564	5139.38	9.11	NA	NA

p-value shown as 0 due to rounding; actual $p < 2.2 \times 10^{-16}$

Table: Table 3. R-squared by transformation

	y	log(y)	1/y	sqrt(y)	1/sqrt(y)
x	0.5610	0.6330	0.6602	0.6026	0.6520
log(x)	0.5626	0.6260	0.6434	0.6002	0.6401
1/x	0.5547	0.6087	0.6167	0.5877	0.6180
sqrt(x)	0.5630	0.6308	0.6530	0.6026	0.6473
x^2	0.5505	0.6302	0.6668	0.5956	0.6538

Table: Table 4. ANOVA, Part B

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
xt	1	23917.69	23917.69	916.98	0
Residuals	472	12311.22	26.08	NA	NA

p-value shown as 0 due to rounding; actual $p < 2.2 \times 10^{-16}$

Table 5. Lack-Of-Fit Test, 1/y on binned x

```
Analysis of Variance Table

Response: yt
              Df  Sum Sq Mean Sq  F value Pr(>F)
x_bin          1 23845.6 23845.6 919.9754 <2e-16
Residuals    472 12383.3    26.2
 Lack of fit  17   589.8    34.7   1.3385 0.1639
 Pure Error  455 11793.5    25.9
```